Introduction

Maximum likelihood is the dominant estimation technique in modern statistics. The idea: given data and a parametric model, choose the parameter values that make the observed data most probable. This intuition supports a powerful general method with well-understood consistency, efficiency, and distributional properties, and it underpins almost every regression model in a statistician's toolkit.

Prerequisites

The reader should know what a probability density is, be comfortable with calculus at the level of derivatives and log transforms, and be able to call `optim()` in R.

Theory

For iid data $X_1, \ldots, X_n$ from a density $f(x \mid \theta)$, the **likelihood function** is

$$L(\theta) = \prod_{i=1}^{n} f(X_i \mid \theta),$$

and the **log-likelihood** is $\ell(\theta) = \sum \log f(X_i \mid \theta)$. The **MLE** is

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} \ell(\theta).$$

When ℓ is smooth, the MLE satisfies the **score equation** $\frac{\partial \ell}{\partial \theta} = 0$. The score is the gradient of the log-likelihood; its expectation under the true parameter is zero.

Under regularity conditions (smoothness, identifiability, interior optimum, finite Fisher information), the MLE is:

- **Consistent**: $\hat{\theta}_n \xrightarrow{P} \theta_0$.
- **Asymptotically normal**: $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1})$.
- **Asymptotically efficient**: its variance attains the Cramer-Rao lower bound.
- **Equivariant**: for any one-to-one transformation g, $g(\hat{\theta}_{\text{MLE}})$ is the MLE of $g(\theta)$.

Standard errors for the MLE come from the inverse of the observed Fisher information, $\mathcal{I}(\hat{\theta}) = -\partial^2 \ell / \partial\theta \partial\theta^\top$, evaluated at $\hat{\theta}$. Confidence intervals can be based on the normal approximation or, better, on the likelihood ratio, which is invariant under reparameterisation.

Assumptions

The asymptotic theory assumes a well-specified model, finite Fisher information, enough smoothness for the Taylor expansion, and a parameter interior to the parameter space. Boundary parameters (variance at zero, proportion at 0 or 1) and non-identifiable models break the standard theory.

R Implementation

```r
set.seed(2026)
x <- rnorm(200, mean = 5, sd = 2)

nll <- function(par) {
  mu    <- par[1]
  sigma <- par[2]
  if (sigma <= 0) return(Inf)
  -sum(dnorm(x, mean = mu, sd = sigma, log = TRUE))
```

```
}

fit <- optim(par = c(0, 1), fn = nll, hessian = TRUE,
             method = "L-BFGS-B", lower = c(-Inf, 1e-6))
mle <- fit$par
se  <- sqrt(diag(solve(fit$hessian)))

data.frame(parameter = c("mu", "sigma"),
           estimate = mle, se = se,
           ci_low = mle - 1.96 * se,
           ci_high = mle + 1.96 * se)
```

The negative log-likelihood is minimised with `optim()`; standard errors come from the inverse Hessian.

Output & Results

```
  parameter estimate    se  ci_low ci_high
1        mu    5.093 0.143   4.812   5.374
2     sigma    2.022 0.101   1.823   2.221
```

The MLE recovers the true parameters ($\mu = 5$, $\sigma = 2$) with uncertainty quantified by the observed-information-based standard errors.

Interpretation

For a manuscript, the MLE is typically reported as "parameters were estimated by maximum likelihood (Fisher-scoring convergence to 1e-8)", with point estimates, standard errors, and either Wald or profile-likelihood confidence intervals. Information criteria (AIC, BIC) are functions of the maximised likelihood and are reported alongside.

Practical Tips

- Always work on the log scale for numerical stability; products of densities underflow quickly.
- Start the optimiser at sensible initial values – method-of-moments estimates are a good default.
- Parameterise so that the parameter space is unbounded (log-variance instead of variance, logit-probability instead of probability); this makes `optim()` easier and avoids boundary issues.
- For models with many parameters, use `nlm()`, `optim(..., method = "BFGS")`, or specialised packages (`bbmle`, `maxLik`, or model-specific tools like `glm()`).
- When the MLE is at a boundary, standard asymptotic inference fails; use bootstrap CIs or specialised boundary-aware procedures instead.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain

- Data types, tidy data, accuracy and precision
- Import, joins, and missingness with dplyr